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Sir Kenelm Digby (1603–1665): diplomat, entrepreneur, privateer, duellist, scientist and philosopher

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Abstract

Kenelm Digby was a man with widely differing interests. He studied at Oxford but left without a degree. At the age of 20, he was involved in a dramatic street brawl in Madrid; the following year saw him as a successful privateer in the Mediterranean and adjacent Atlantic waters, and he was then appointed a naval administrator with lucrative trade monopolies. He practised medicine without formal qualifications and gained notoriety by promoting the use of 'sympathetic' powder for healing wounds. Befriended by King James I and his son who succeeded him, he became chancellor to Queen Henrietta and was knighted by King James I on 21 October 1623 and appointed to the Privy Council. At one time, he even spent a brief time in prison, but he later became a founding member of the Royal Society and published two major philosophical treatises as well as a popular cookbook. He designed and manufactured wine bottles with tapered necks for simpler pouring and square sides for easy stacking. He dabbled in mathematics and corresponded with Fermat of 'Last Theorem' fame. In 1641, he challenged a French nobleman who publicly cast a slur on the British King to a 'digladiation' (a sword duel) and quickly dispatched him. He was a friend and admirer of Sir Thomas Browne whose 'Religio Medici' he helped to publish. The Cowlishaw Library holds a 1659 5th edition copy of this remarkable book, with Digby's name on the spine. Truly, Digby must qualify as a Renaissance man.

Introduction

Sir Kenelm Digby is not so widely known as Sir Thomas Browne is, the highly regarded 17th century English physician, a man whose opinion was frequently sought by colleagues, a devoted Christian, widely acclaimed for his published work 'Religio Medici' which is still in print after 350 years. But it was Digby who assisted Browne to have the definitive copy of his book published in 1642 in London. A 1659 copy of this book resides in the Cowlishaw Library of the Royal Australasian College of Surgeons and it includes 'Observations' by Digby in language typical during that time (Fig. 1).

Kenelm Digby was certainly remarkable – he possessed a variety of talents which he used in an astonishing number of diverse activities packed into his 62 years. His interests were so varied and successful that he might well be compared favourably to earlier Renaissance greats.

Background

Kenelm was born in England on 11 January 1603, 3 months after the death of Queen Elizabeth I, at Gayhurst in Buckinghamshire to Sir Everard Digby and his wife Mary. In 1605, Sir Everard, a staunch Catholic convert, was one of a group of conspirators led by Robert Gatesby who planned to blow up the English Houses of Parliament. Guy Fawkes, whose name came to identify the plot, actually planted the explosives, but he was arrested on 5 November and the plot failed. Gatesby was shot while resisting arrest and the other conspirators were executed, including Sir Everard, who was beheaded in St Paul's Churchyard on 31 January 1606.

For his own safety, the young Kenelm, then aged 3, was separated from his Catholic mother, and, for a time, became the ward of William Laud, who himself had a remarkable career, becoming Privy Councillor in 1626, Bishop of London in 1628 and Archbishop of Canterbury in 1633. Laud was friendly with both James I and Charles I, and under the patronage of the latter, he endeavoured to restore the tall wooden spire which had caught fire and toppled from the tower of the massive mediaeval St Paul's Cathedral in 1561. Laud never completed this work, and because of his extensive involvement in Church politics, he was impeached by the Long Parliament in 1640 and summarily executed.

After Laud's guardianship, Digby became the ward of his uncle, Sir John Digby, who was appointed Ambassador in Madrid and later

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Fig. 1. Religio Medici – 1659 (Cowlishaw Library, RACS, Melbourne).

became Earl of Bristol. His father's execution and the automatic forfeiture of his estate was followed by a litigation on behalf of the young son which resulted in some of the unconfiscated property being remitted to the lad, and this earned him 15 000 pounds per year. He stayed with his uncle in Spain for 6 months until April 1618 and quickly developed a familiarity with court and diplomatic affairs, and at the age of 15, he entered Oxford's Gloucester Hall (later Worcester College). He remained there until 1620 and studied Physical Science and Mathematics under Thomas Allen who was so impressed by his pupil's brilliance that he bequeathed many of his valuable books and manuscripts to him, and Digby later presented them to the Bodleian Library.

For the next 3 years, Digby travelled widely throughout Europe, a popular custom for young gentlemen at that time, and he returned to England in September 1623 where he was befriended by Prince Charles, and surprisingly, for one so young, he was knighted by King James I on 23 October of that year. In 1625, he secretly married Sir Edward Stanley's beautiful daughter Venetia whose antenuptial reputation has been described as 'not spotless', but their marriage was a happy one, and their union produced four sons and a daughter. Sadly, Venetia died suddenly in 1633, possibly from tuberculosis.

Digby's medical activities

Back in London in 1623, after his European sojourn, Digby practised as a 'quack'. Conventional physicians of the time regarded him as a non-qualified practitioner, but he enjoyed a successful following in England. When he was a courtier in Spain, he studied the lip reading technique of a Catholic padre who found a way to teach deaf mutes to speak. But he gained widespread recognition on the continent as well as at home by his promotion of the 'powder of sympathy' in wound management, after he presented an account of its use and a suggested mode of its action to the Academy of Sciences at the University of Montpellier in France in 1657 (Fig. 2). Digby was there for treatment of his bladder stone, and his 'discourse' on the subject was published in Paris and in London in 1658 and was reprinted more than 20 times. The technique was to mix a salve of ferrous sulphate with discharge from a wound, and then the salve

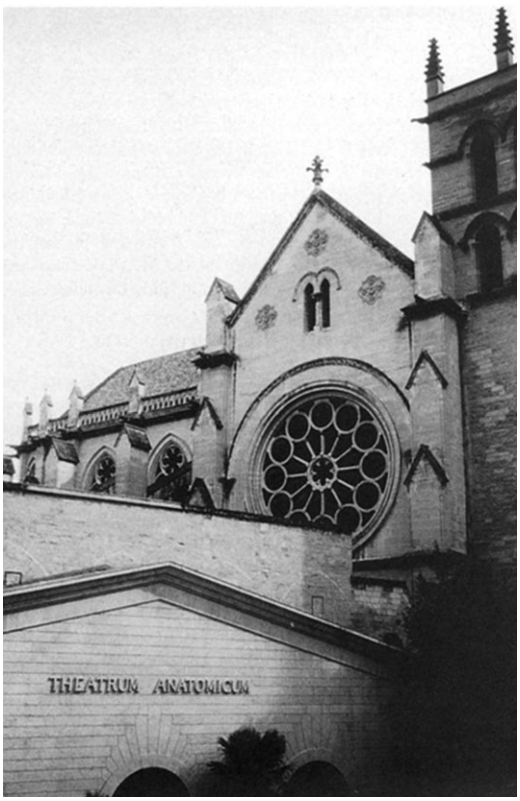

Fig. 2. Anatomy Theatre Montpellier University (photograph by the author).

was applied to the causative agent, sword or gun, and the wound itself was simply cleaned without the usual application of caustics or cautery or other irritants, the same gentle form of wound treatment promoted by Ambrose Paré over 100 years before. Digby considered that the 'powder' worked by the transfer of 'blood atoms' from the wound coming into contact with 'curative atoms' in the salve and these were then attracted back to the wound where they effected the cure, whereas in reality, the unhindered 'vis medicatrix naturae' of the blandly treated wound was allowed to operate successfully.

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Digby – Master of digladiation

Duelling with swords has a long history, but it was not until 1656 that Digby's contemporary and noted logophile Sir Thomas Browne coined the descriptive term 'digladiation' for such encounters. In rare use today, like duelling itself, the word is attractive and rolls smoothly over the tongue and its disappearance would be regrettable.

During his time in Europe from 1620, Digby first spent a short time in France; then he moved to Italy where he quickly learned the language and became an expert fencer. His skill in this activity must have been highly regarded, for a 1621 Italian publication on fencing, 'Oplomachia', was dedicated to him. This particular expertise stood him in good stead on at least two occasions.

In the summer of 1623, he visited his uncle, by then Earl of Bristol, in Madrid. After dinner with him one night, Digby set off to walk back to his lodgings accompanied by his cousin and a friend. On the way, the three men were attacked without warning by a group of 15 'armed' men and a fierce sword fight followed. Digby relates in his 'Memoirs' how the attackers were overcome, with several of them wounded, and their leader slain by himself; after which success, he was acclaimed a hero by the local population.

In 1641, Digby was present at a banquet in Paris where he felt that a French nobleman, Mont le Ros, spoke disparagingly about King Charles I. Like Nelson 150 years later, Digby must have felt the need '. . . to consider every man your enemy who speaks ill of your King', for he promptly challenged the man to a duel, and in the ensuing encounter, he killed his opponent with a thrust through the heart. However, the offence was pardoned by the French King on the grounds that Digby was defending the fair name of his King and Digby was given safe conduct out of France.

On his return to England and despite his popularity and hero status, he soon found himself in prison presumably because of his well-known and active opposition to the spread of Puritanism.

Digby's sea adventures

By the year 1627, much of England's maritime fleet was at the mercy of pirates, principally armed merchant vessels flying French, Dutch, Flemish or Spanish flags. Digby apparently decided that retaliation was called for; he equipped two ships with a full complement of sailors and, with King Charles' knowledge, set sail from Deal in December of that year. Within a month, he had captured several Flemish and Dutch vessels near Gibraltar. Because of sickness in his crew, possibly scurvy, he then anchored near Majorca until the men had recovered, and then promptly captured another well-laden Dutch vessel.

There were by now more frequent attacks on English ships in the Eastern Mediterranean near the Turkish coast. This prompted Digby to move to those waters and on 11 June 1628, he unleashed a fierce attack on unsuspecting French and Venetian vessels anchored in the Turkish harbour of Scanderoon (now Alexandretta) (Fig. 3). After overcoming fierce resistance, he claimed considerable booty and returned to England in February 1629 to a warm welcome from the court as well as the general public. He was appointed a naval administrator, then Governor of Trinity House, the Lighthouse Authority for England and Wales, the Channel Islands and Gibraltar; and additionally, he received the Patents of a Monopoly of Trade for Sealing Wax in Wales and the adjacent border counties; and these positions produced a significant additional income.

Fig. 3. Arrow showing Gulf at Scanderoon from 1845 (map in possession of the author).

For the next few years, he spent time with his family and pursued many activities at home, but he was greatly saddened by the sudden death of his beloved wife Venetia on May 1633, possibly from tuberculosis; and he promptly arranged for van Dyck to paint her shortly after her death, exactly as she was found. Along with praise from other notables, Ben Jonson wrote a poem in her honour.

Digby's versatility

Mathematics

Pierre de Fermat (1601–1665), a contemporary of Digby, qualified as a lawyer in Paris but soon became interested in Mathematics, Numbers Theory and Geometry as a hobby; he devised a great number of theorems and also developed a form of Calculus a few years before Newton. Around 1637, he wrote in the margin of his copy of the 'Arithmetica' written by the great mathematician, Bioplantus in Alexandria in the 3rd century. '. . . It is impossible to separate a cube into two cubes or a biquadrate into two biquadrates or generally any power except a square into 2 powers with the same exponent. I have discovered a truly marvellous proof of this which however, the margin is not large enough to contain'. This theorem can be expressed algebraically as

$$x^n + y^n = z^n$$

where n can never be greater than 2. Many of Fermat's propositions were either proved or disproved over the next 200 years, but despite many attempts, this particular theorem remained unsolved and it became known as 'Fermat's Last Theorem'. It was in fact only solved by the English mathematician, Andrew Wiles in 1994. Digby apparently corresponded with Fermat, in particular about Numbers Theory and Pythagorean Triangles; and in 1655, the Englishman John Wallis (1616–1703) published his 'Arithmetica Infinitorum' in which he, too, anticipated calculus, and he fulsomely dedicated the work to Digby, who obviously had impressed a talented French and an English mathematician.

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Domestic interests

Digby collected numerous medicinal and culinary recipes which included varieties of mead, advice on roasting wild boars and natural remedies for such conditions as aches, mad dog bites and agues. A certain George Hartman, sometime Digby's steward, collected his master's recipes and many were published posthumously between 1668 and 1682 and they proved extremely popular.

Commerce

In the 1630s, Digby owned glassworks where he produced wine bottles with high tapered necks which made pouring easier. His glass proved to be stronger than most and was darker in colour and therefore let in less light, so protecting the contents. The bottles themselves were square and easy to stack, and in 1662, the English Parliament formally recognized the design as his, thereby rebutting other claimants.

Philosophy and science

Digby had been imprisoned in Winchester House for almost 2 years up until July 1643 and he was in and out of England several times after that, once being banished. He published a Folio volume of 510 pages touching on a variety of subjects and containing two philosophical treatises, 'The Nature of Bodies' and the 'Immortality of Reasonable Souls'. These included descriptions of chicken embryology based on the study of hatching eggs; observations on the circulation of blood and Harvey's seminal work which Digby strongly and publicly supported; and reflex conditioning of animal behaviour. Certainly, he was greatly influenced by the scientific and experimental fervour of the time.

The royal society

By the time of the restoration in 1660, he was finally settled at home after his sojourn in Montpellier, and he became closely associated with notable scientists like Wallis, Hooke and Boyle, and he argued strongly for the formation of a formal scientific body. On 23 January 1660, at Gresham College, he presented a paper entitled 'A Discourse Concerning the Vegetation of Plants' to members of the Society for Promoting Philosophical Knowledge by experiments and this became the first publication of that Society in 1661 (Fig. 4). In this paper, he developed the view that plants needed air for life as well as soil nutrients and water. Digby became a founding member of the Royal Society which was incorporated in 1663 and he was appointed a member of its first Council.

Finis

This remarkable man died of 'stone' on 11 June 1665, ironically his birth date and the anniversary of the sea battle he had fought near Turkey when he was only 25. He was buried alongside his wife at Christchurch, Newgate, close by St Paul's, just 15 months before that area was decimated by the Great Fire. His epitaph by Richard Farrar says it all.

Under this Tomb the Matchless Digby lies;
Digby the Great, the Valliant, and the wise;
This Ages Wonder for His Noble Parts;
Skill'd in Six tongues, and Learn'd in All the Arts.

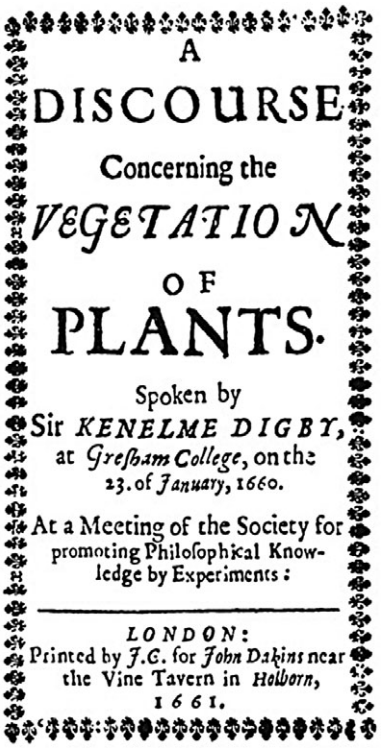
A
DISCOURSE
Concerning the
VEGETATION
OF
PLANTS.
Spoken by
Sir KENELME DIGBY,
at Gresham College, on the
23. of January, 1660.
At a Meeting of the Society for
promoting Philosophical Knowledge by Experiments:
LONDON:
Printed by J.G. for John Dakins near
the Vine Tavern in Holborn,
1661.

Fig. 4. Title page from Digby's *A Discourse Concerning the Vegetation of Plants* (Cowlishaw Library, RACS).

Born on the Day He Dy'd, Th'Eleventh of June,
And that Day Bravely Fought at Scanderdoun.
'Tis Rare, that one and the same Day should be
His Day of Birth, of Death, and Victory.

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